

LangChain for LLMs

All About LangChain
Chapter 4



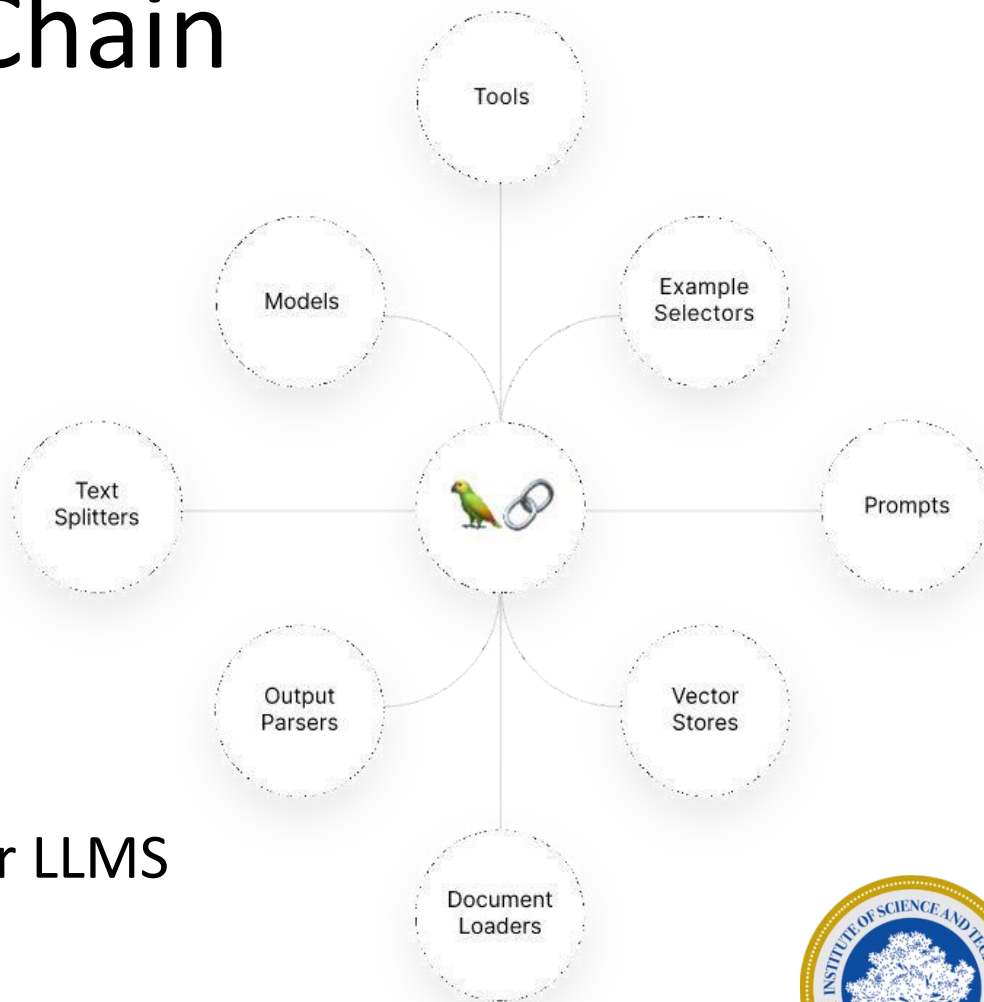
LangChain

- Numpy and Pandas for LLMs, greatly increasing their usability and functionality
 - Connect a language model to other sources of data
 - Allow a language model to interact with its environment



LangChain

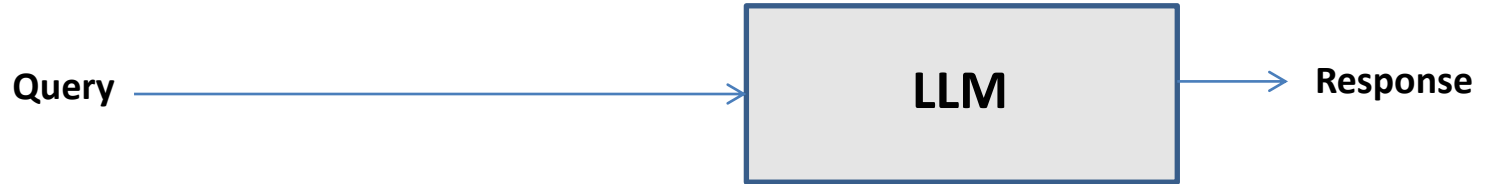
- Python (also JS/TS) framework
- Building blocks
- Swappable components
- Examples
- From PoC to Production
- Speed of improvement



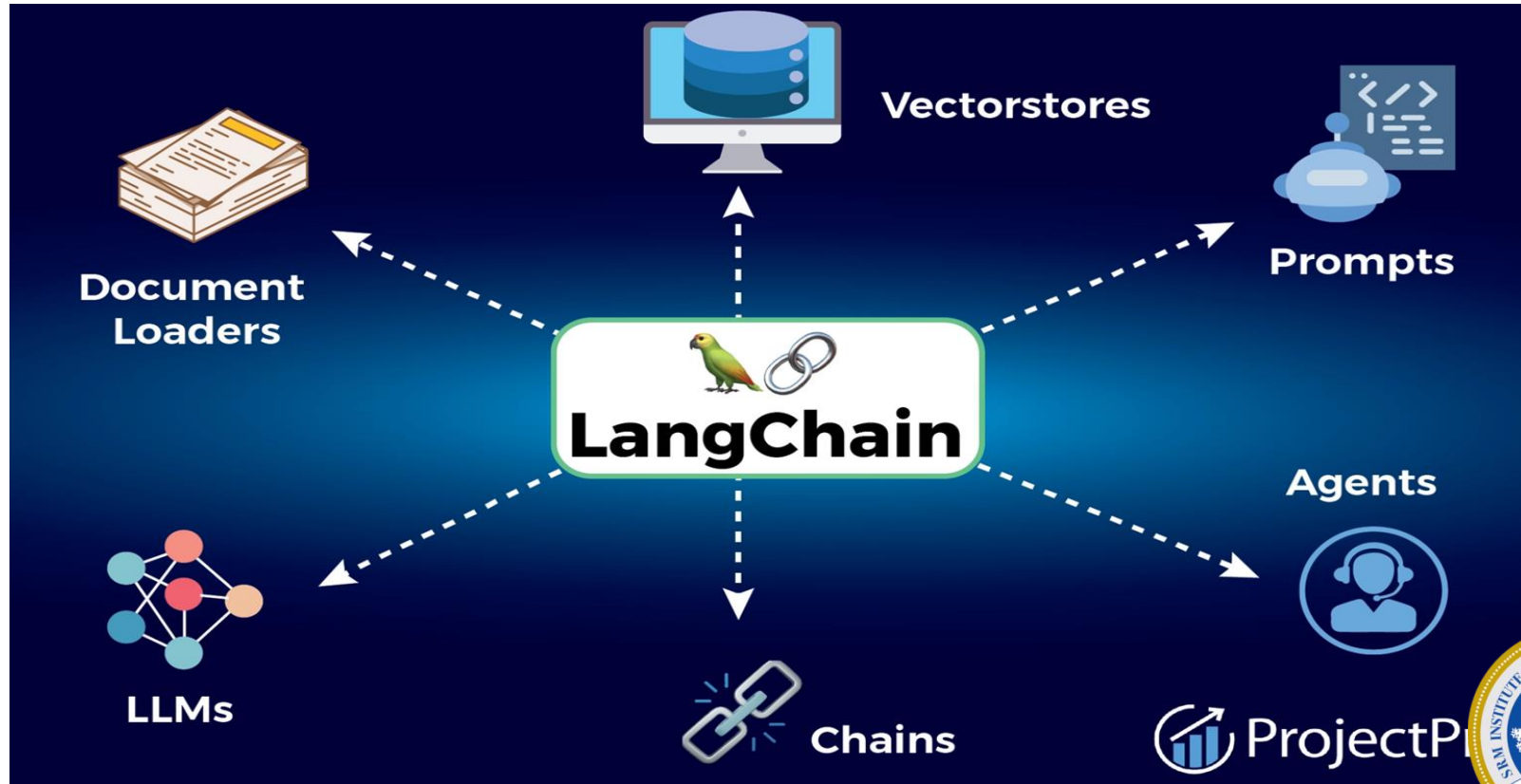
Its basically Numpy/Pandas for LLMS



A Basic Flow



Components of LangChain



Retrieval Augmented Generation (RAG)

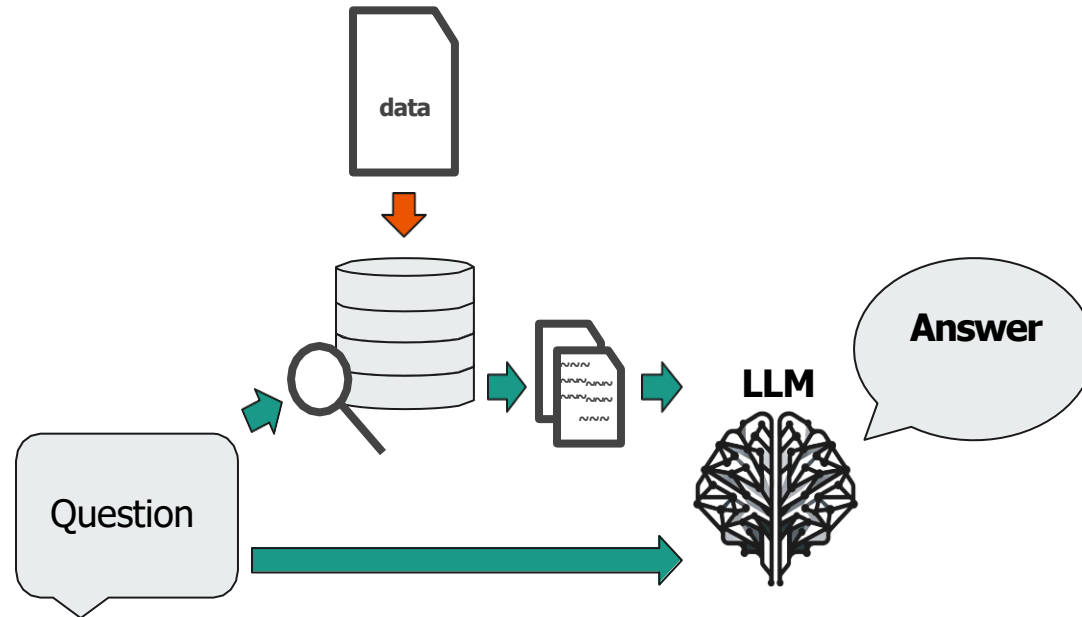


Augment LLM knowledge using additional data

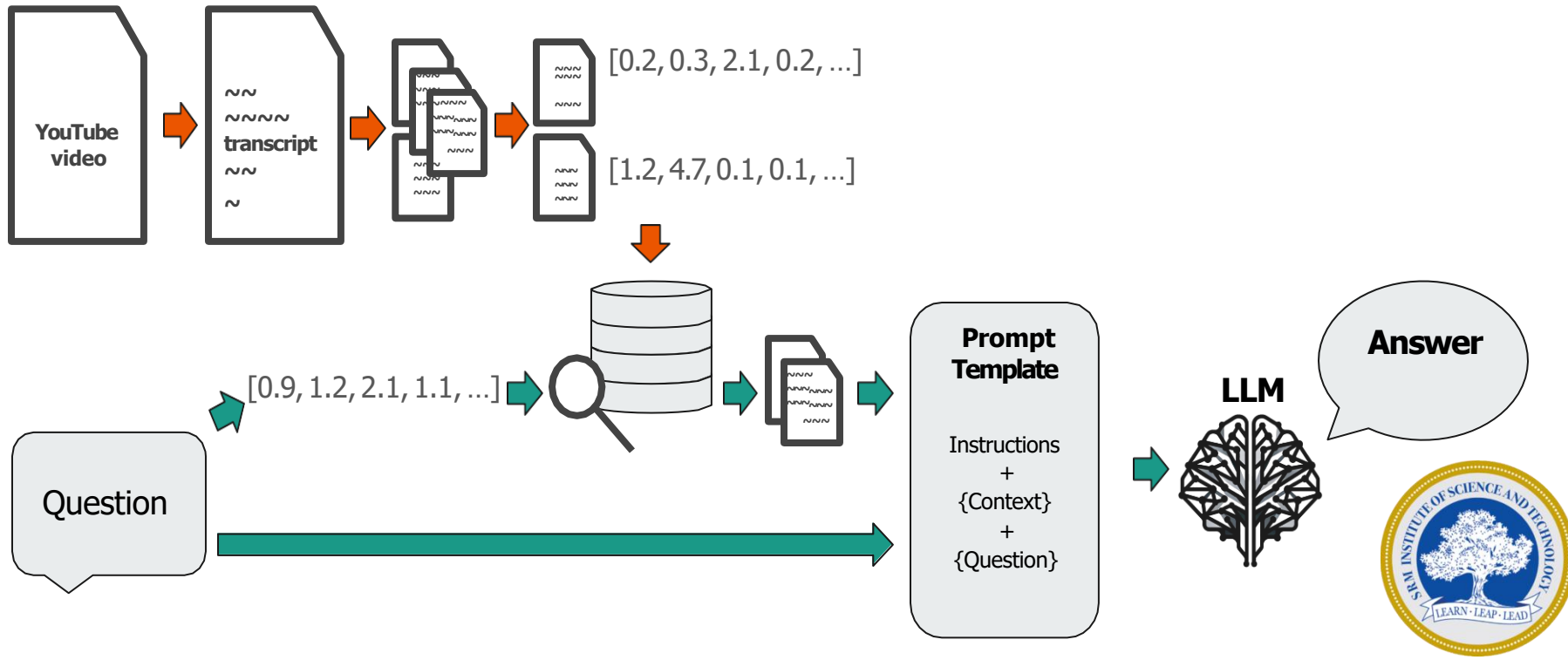
- Combines retrieval +generation
- Data not in training dataset
 - Private data
 - Data after cutoff date, even real time
- Improves accuracy and relevancy
- Supports evidence-Based Responses, can reference source



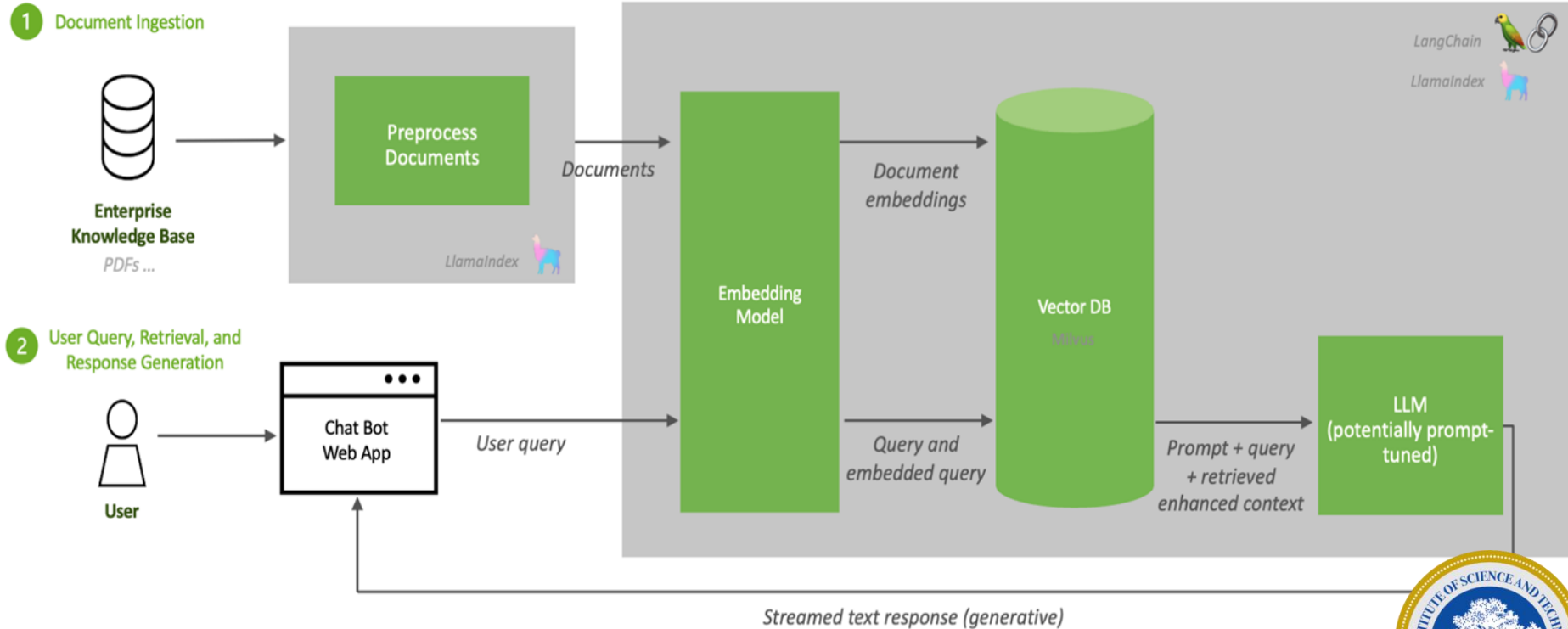
RAG



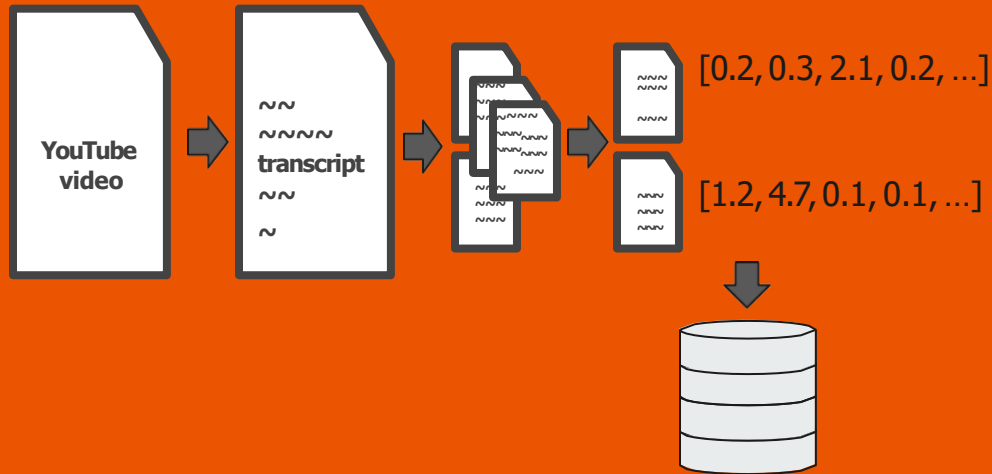
RAG



Retrieval Augmented Generation (RAG) Sequence Diagram

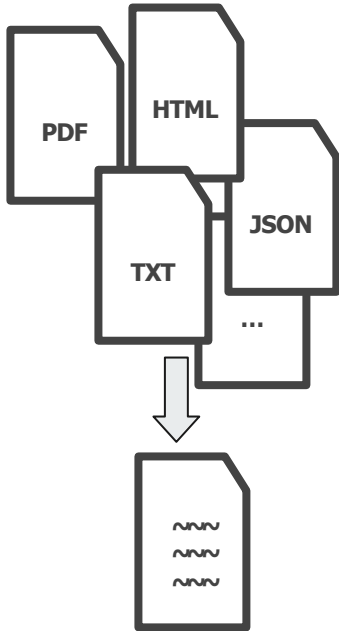


Preparing and storing data

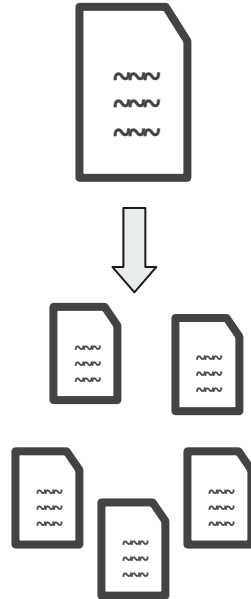


RAG

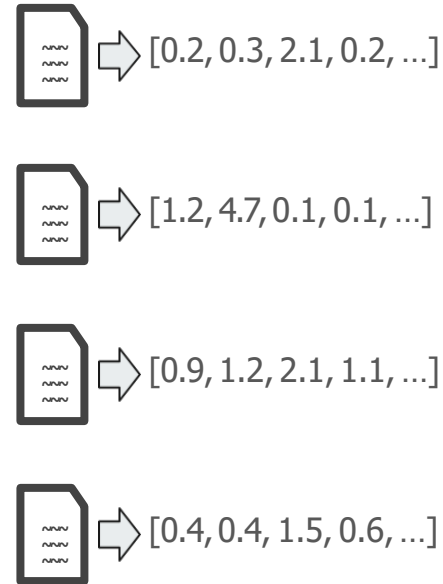
Document loader



Text Splitter

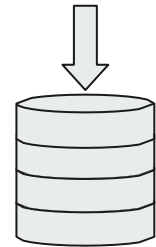


Embedding Function

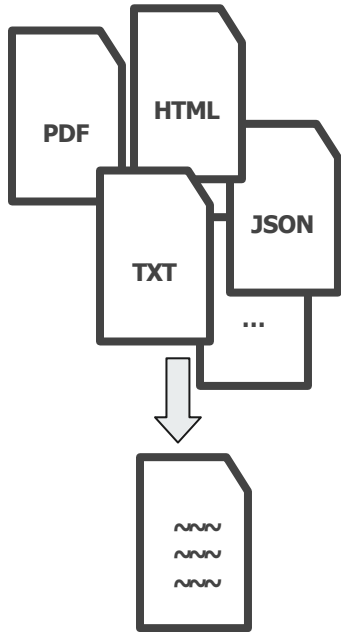


Vectorstore

$[0.2, 0.3, 2.1, 0.2, \dots]$
 $[1.2, 4.7, 0.1, 0.1, \dots]$
 $[0.9, 1.2, 2.1, 1.1, \dots]$
 $[0.4, 0.4, 1.5, 0.6, \dots]$



Document Loaders



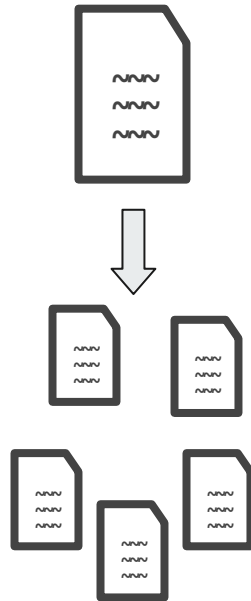
Arxiv
CSV
Discord
Email
EPub
EverNote
Facebook Chat
Figma
Git
GitHub
HTML
JSON
Markdown
Mastodon
MediaWiki Dump

Microsoft Word
MongoDB
Open Document Format (ODT)
Pandas DataFrame
PubMed
ReadTheDocs Documentation
Reddit
RSS Feeds
Slack
Snowflake
Telegram
X
URL
WhatsApp Chat
Wikipedia
XML
YouTube audio
YouTube transcripts



Text Splitters

Break text into smaller chunks



What is FOSDEM?

FOSDEM is a free and non-commercial event organised by the community for the community. The goal is to provide free and open source software developers and communities a place to meet to:

- get in touch with other developers and projects;
- be informed about the latest developments in the free software world;
- be informed about the latest developments in the open source world;
- attend interesting talks and presentations on various topics by project leaders and committers;
- to promote the development and benefits of free software and open source solutions.

Participation and attendance is totally free, though the organisers gratefully accept donations and sponsorship.## Developer rooms

The FOSDEM team feels it is very important for free and open source software developers around the world to be able to meet in "real life".

To this end, we have set up developer rooms (devrooms) with network/internet connectivity and projectors where teams can meet and showcase their projects. Devrooms are a place for teams to discuss, hack and publicly present latest directions, lightning talks, news and discussions. We believe developers can benefit a lot from these meetings.## A bit of history

In 2000, Raphael Bauduin, a fan of the Linux movement in Belgium, decided to organise a small meeting for developers of Open Source software. He called it 'Open Source Developers' European Meeting' (OSDEM).

Raphael created a mailing list, a small website and spread the word to people around him. Only a few weeks later, lots of people were waiting for an exciting event in Brussels! Invitations were sent to well-known figures in the community: Rasterman, Fyodor, Jeremy Allison and so on. They all gave a very positive response, and OSDEM was on the road to success.

For the second year, OSDEM was renamed FOSDEM. And now, many years later, it has grown into the event it is today. We now try to cover a wide spectrum of free and open source software projects, and offer a platform for people to collaborate. Every year, we host more than 5000 developers at the ULB Solbosch campus. Raphael is no longer the driving force behind FOSDEM. After 7 years of hard work he left the team for new Open Source plans. The FOSDEM flag is now proudly carried by the following people:



Embeddings

- Numerical representation
- **Vectors** in **High-dimensional space**
- Each dimension reflects an aspect
- Similarity = Proximity in embedding space



⇒ [0.2, 0.3, 2.1, 0.2, ...]



⇒ [1.2, 4.7, 0.1, 0.1, ...]



⇒ [0.9, 1.2, 2.1, 1.1, ...]



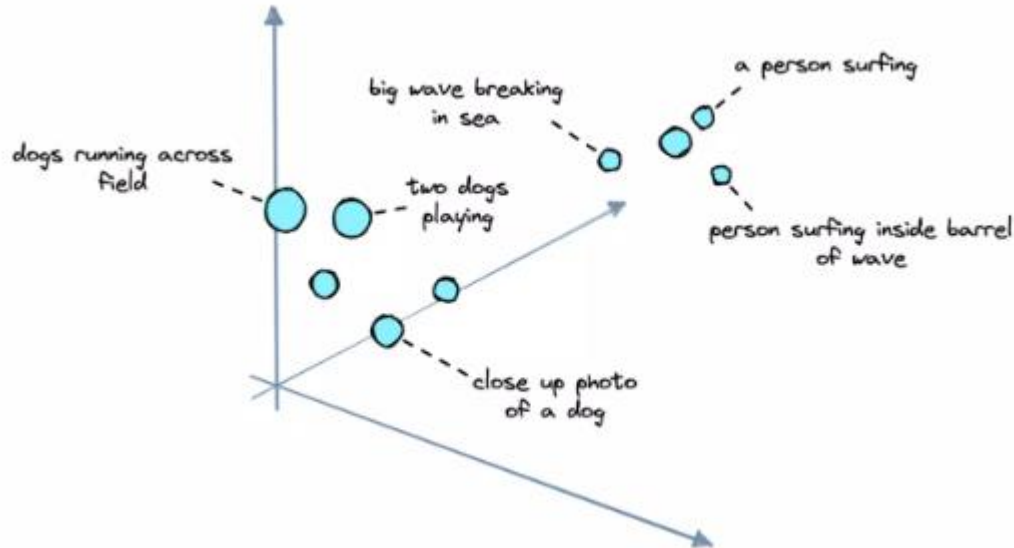
⇒ [0.4, 0.4, 1.5, 0.6, ...]



Vector DB

Storing embeddings

- Stores
- Search
- Retrieve

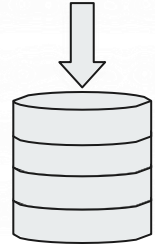


[0.2, 0.3, 2.1, 0.2, ...]

[1.2, 4.7, 0.1, 0.1, ...]

[0.9, 1.2, 2.1, 1.1, ...]

[0.4, 0.4, 1.5, 0.6, ...]



Vector DB

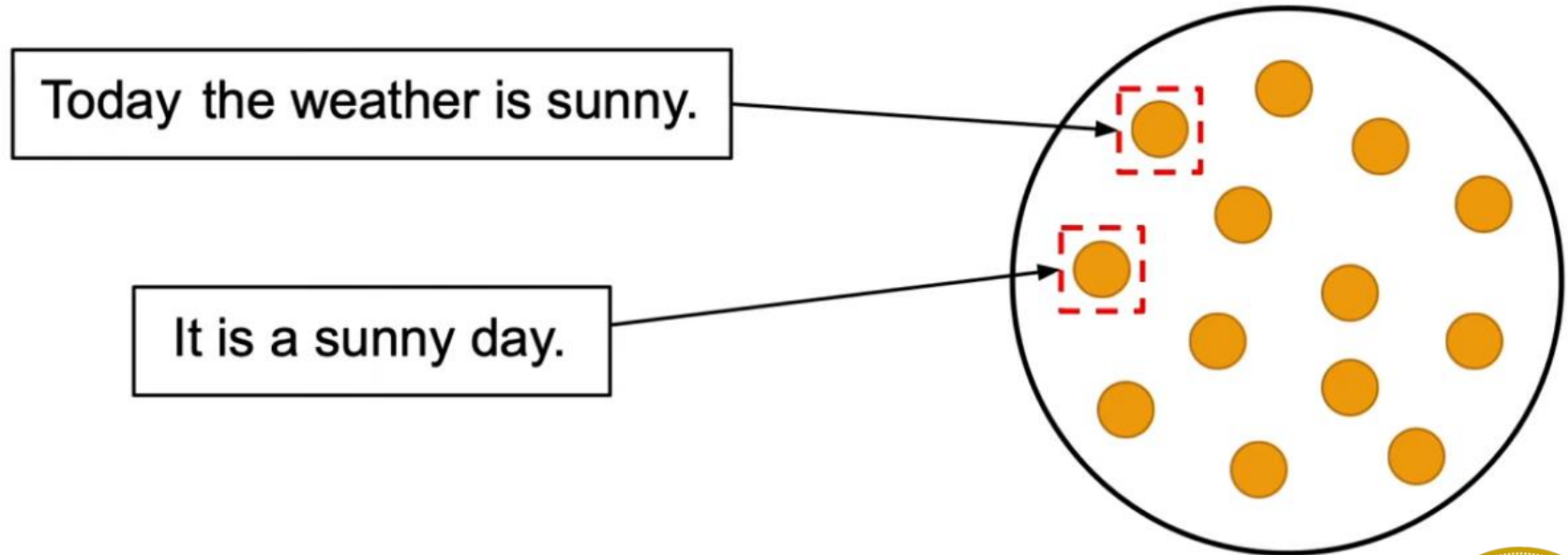


Vector Databases: The Super-Fast Library for Vectors

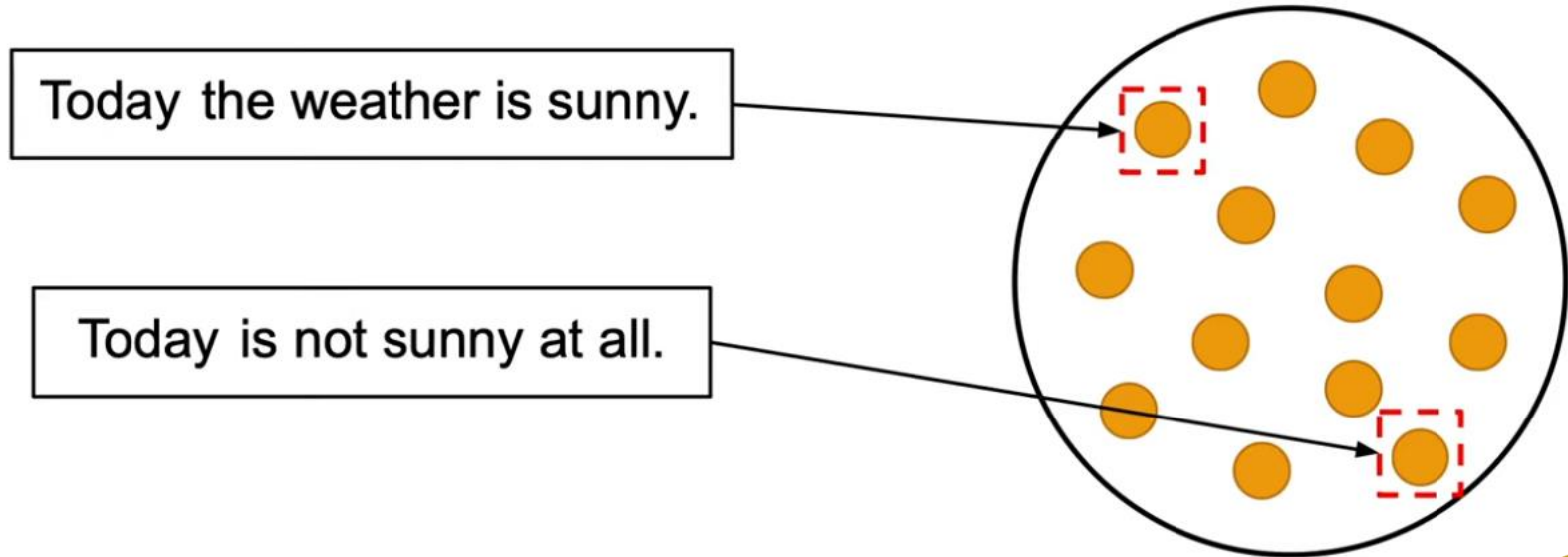
- **Analogy:** Imagine a giant library with millions of books.
- **A normal database (slow way):** To find a book about "ancient Roman history," you'd have to read the title of every single book in the library.
- **A vector database (fast way):** The librarian has already organized the books into sections based on their topic. All history books are in one wing, all science in another. When you ask for "ancient Roman history," the librarian doesn't check the science section at all. They go directly to the history wing and then to the "classical history" aisle to find the closest matches.



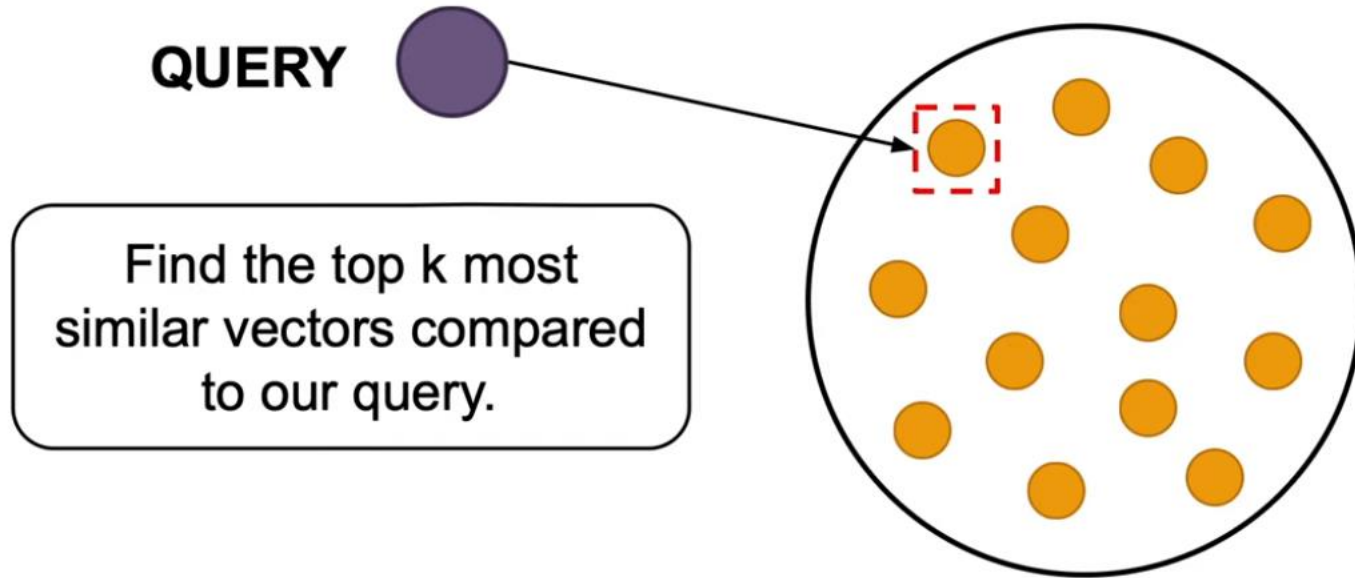
Hierarchical Navigable Small Worlds (HNSW)



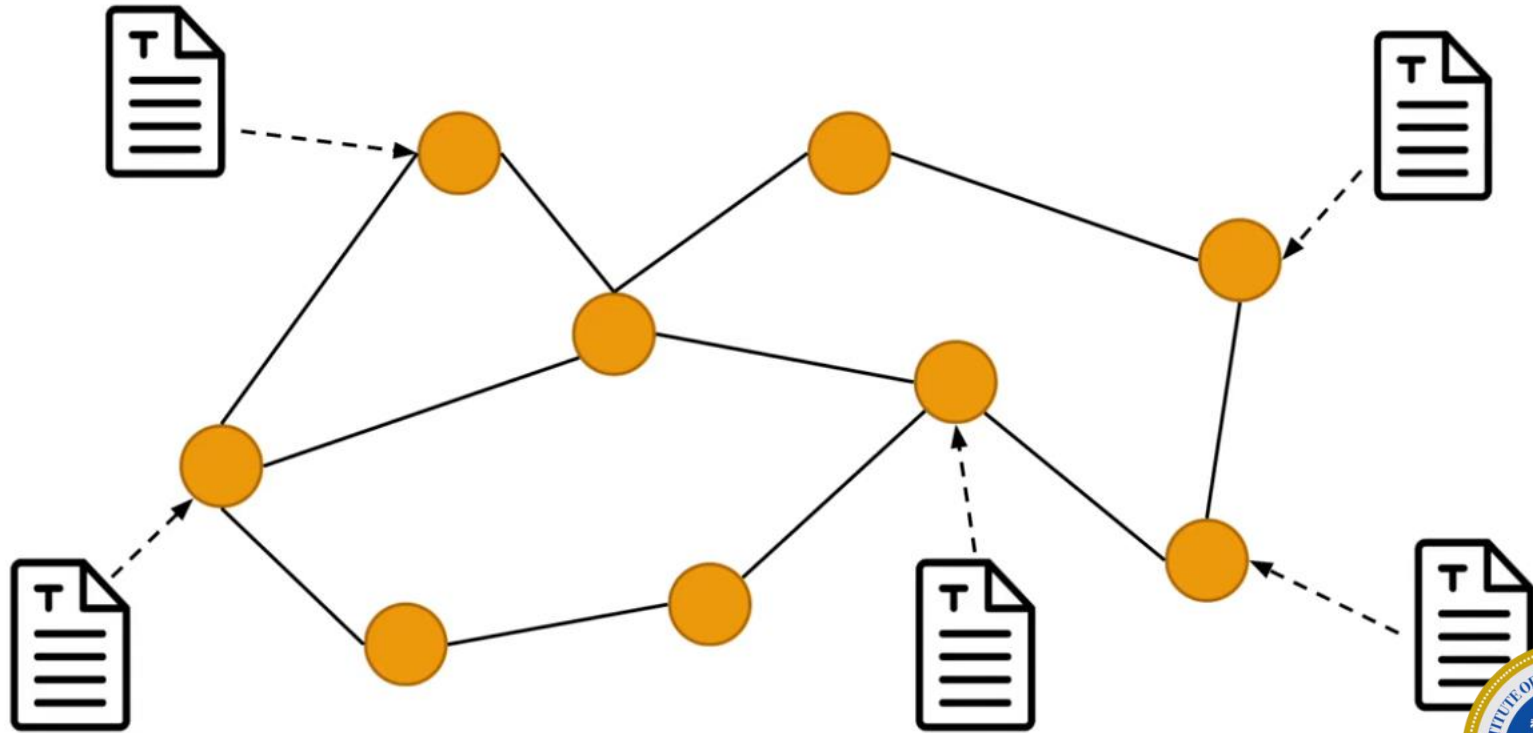
Hierarchical Navigable Small Worlds (HNSW)



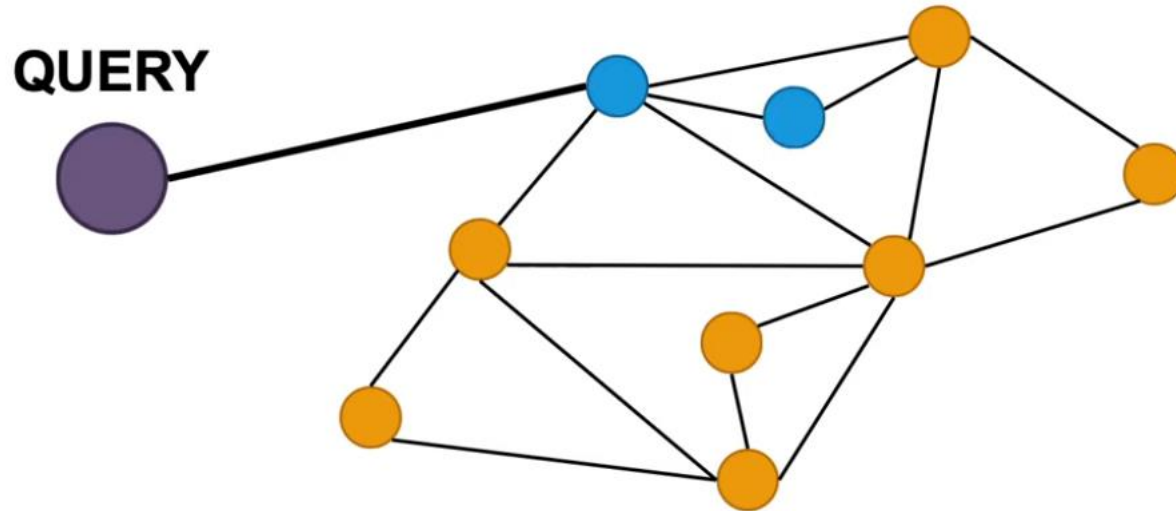
Hierarchical Navigable Small Worlds (HNSW)



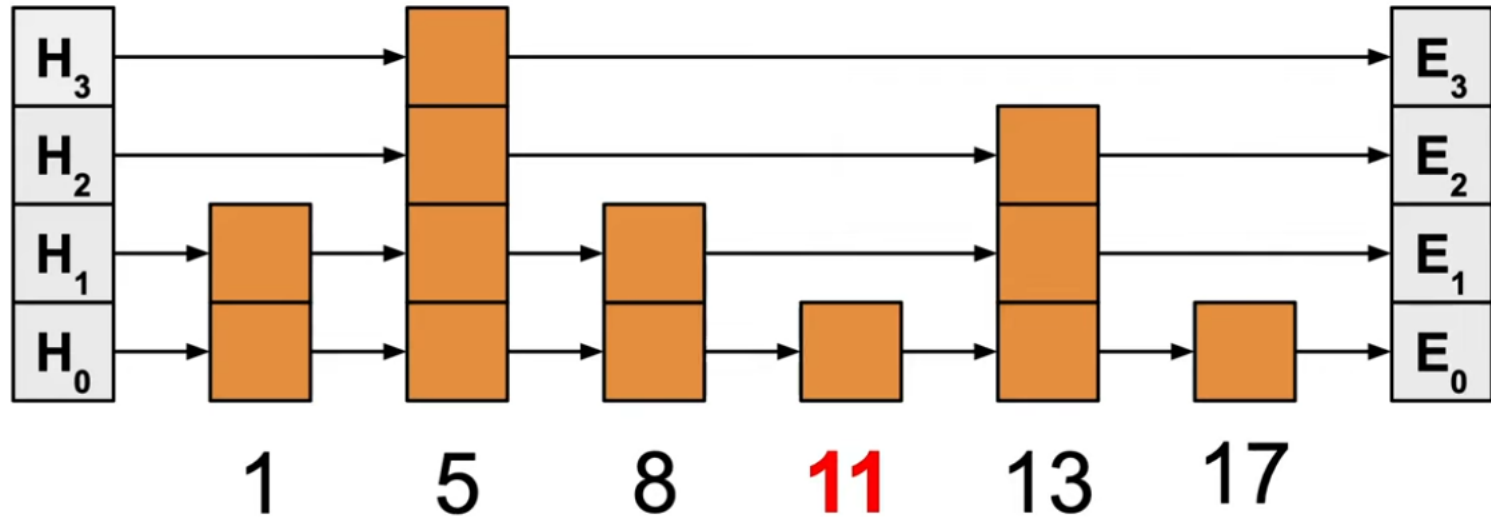
Hierarchical Navigable Small Worlds (HNSW)



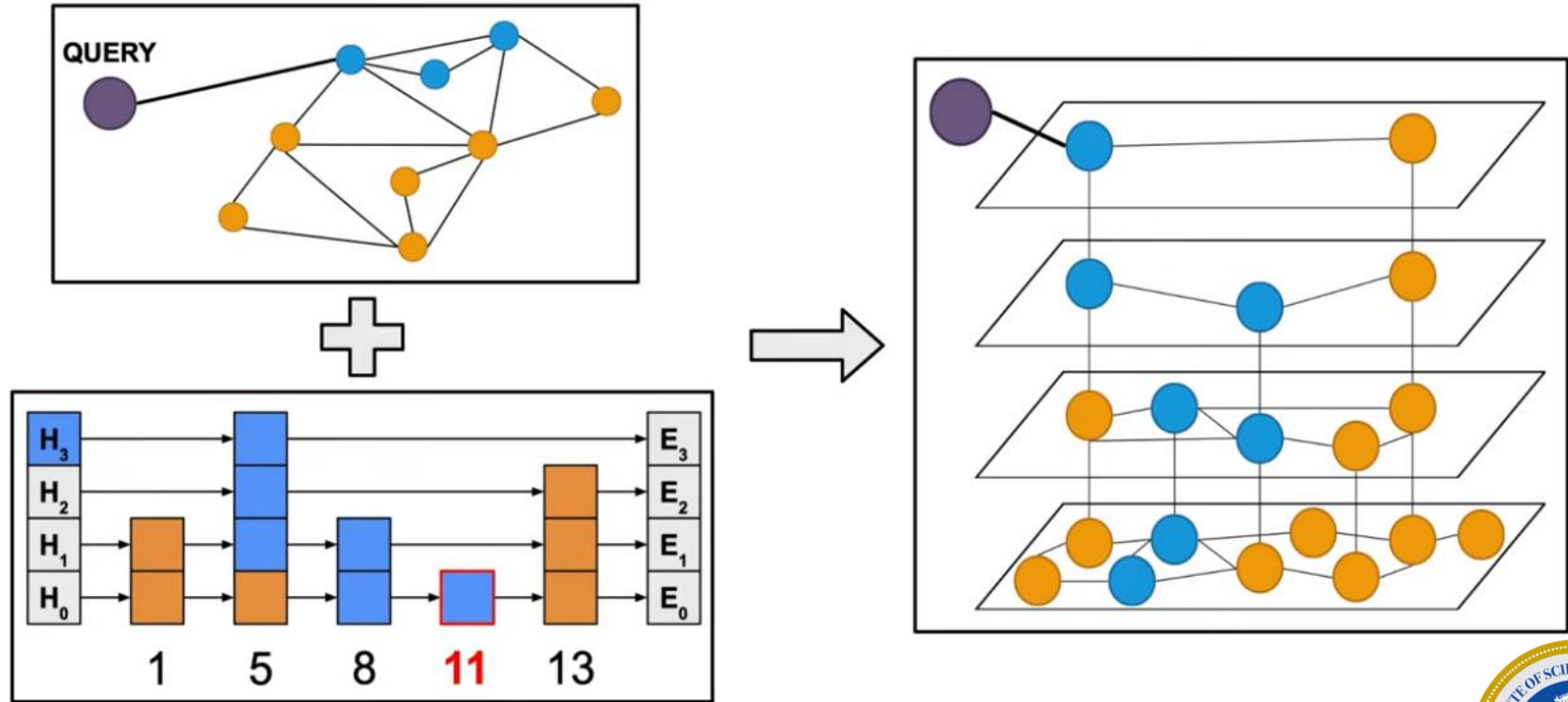
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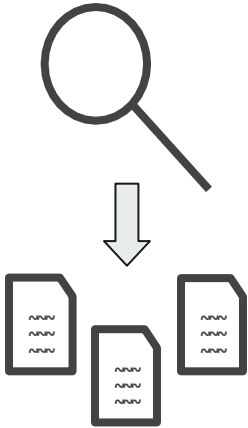


Most Used Vectorstores

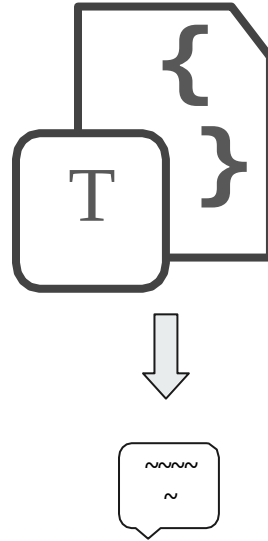


RAG

Retriever



Prompt/Template



LLM



Chain





AI AGENTS TAKES THE SHOW



Generative AI vs. Agentic AI vs. AI Agents



Generative AI

Generative AI generates new content such as text, images, and music based on training data



Agentic AI

Agentic AI performs tasks autonomously and can pursue goals by interacting with its environment



AI Agents

AI agents combine generative and agentic capabilities to accomplish complex objectives



Key characteristics of AI agents

- **Autonomy:** They can operate independently, making decisions and taking action without constant human intervention.
- **Adaptability:** They can learn from data and adapt their behavior based on new information or changing circumstances.
- **Interaction:** They can interact with the environment, other agents, and users through various interfaces, such as text, voice, or visual inputs
- **Goal-Oriented:** They are typically designed to achieve specific goals or objectives, whether it's playing a game, providing customer support, or analyzing data.



Components of an AI Agent

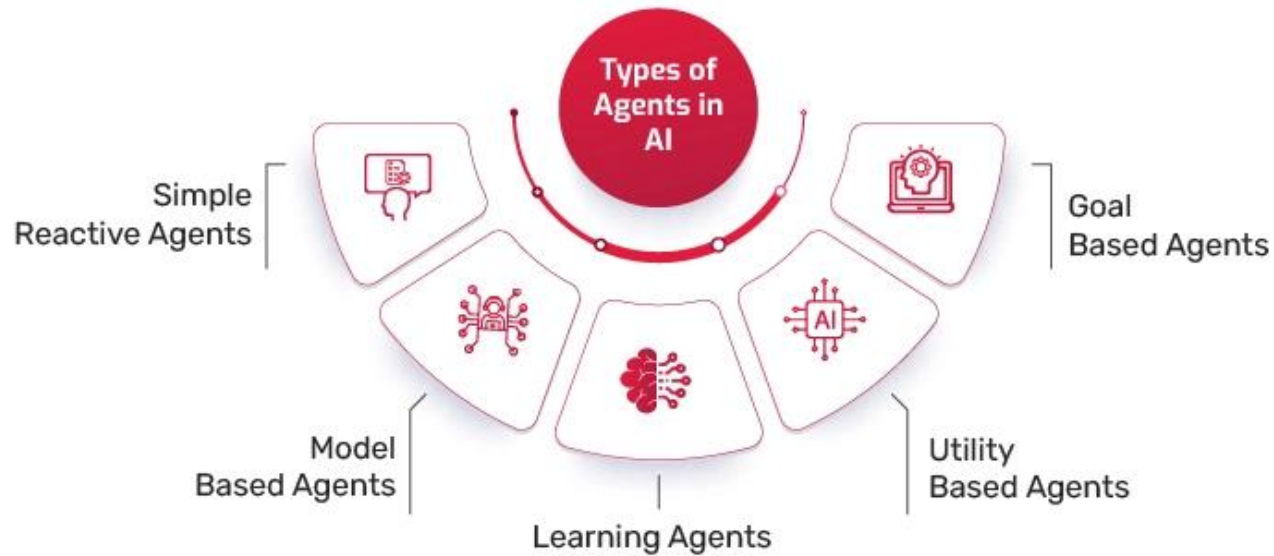
Component	What It Does	Example/Role
Perception	Gathers information from the environment	Sensors, cameras, user input
Reasoning	Processes information to make decisions	Planning, problem-solving, inference
Learning	Improves performance based on experience	Adjusting behavior, updating models
Action	Executes tasks or commands	Moving a robot arm, sending a message
Memory	Stores information about past states	Keeps history for better decisions



Terminologies

Concept	What It Is	What It Gives / Enables	Simple Example
AI Agent	An autonomous entity that perceives, decides, and acts in an environment	Independent decision-making and task execution	Chatbot answering user questions
Agentic AI	AI systems with goal-directed, autonomous behavior and planning abilities	Flexible, scalable AI that can act proactively and adapt	Virtual assistant that schedules your day
Workflow	A defined sequence of tasks or steps to accomplish a goal	Structure and order to complete complex processes	Steps to onboard a new employee
Orchestration	Coordinating multiple agents or processes to work together	Collaboration, task division, error handling, efficiency	Multiple AI agents managing software dev

Types of agents

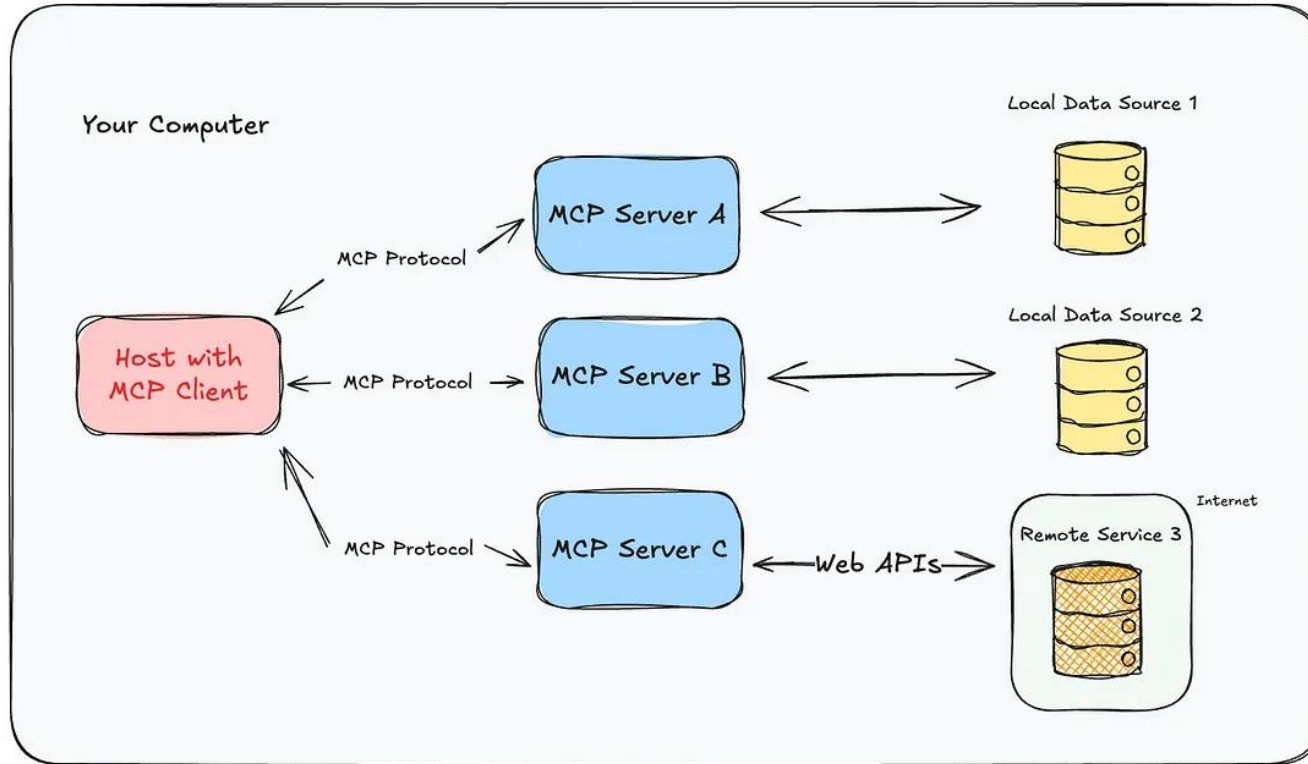


Types of Agent

Agent Type	Why Needed	Simple Example
Reactive	Quick, simple reactions	Light switch
Model-Based	Keeps memory of environment	Maze navigation
Learning	Learns and adapts over time	Spam filter
Utility-Based	Chooses best option among many	Self-driving car balance
Goal-Based	Plans actions to achieve goals	Cleaning robot



MCP



MCP

Traditional APIs VS Model Context Protocol

Aspect	Traditional APIs	MCP
Setup	Manual, one by one	One standard for all tools
Flexibility	Fixed, tool-specific	Dynamic and adaptable
Reuse	Hard to reuse across agents	Easy to reuse everywhere
Scalability	Breaks as systems grow	Built to scale
Agent Compatibility	Needs custom logic per tool	Works out of the box with schema
Tool Discovery	Manual configuration	Automatic, real-time

@elisowski



THANK YOU

